

Sid Mathur's Thematic ETF Screener – Daily Report

2026-07-21

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Action Center

Run date: 2026-07-21 **Universe:** 660 ETFs / 26 clusters

Full PC1: FLAT **Clusters long:** 7/26 **ETFs long:** 191/660 (29%)

Today's events (cluster level)

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

BUY (1)

cluster	theme	spread_buy	p_h5	nss
cluster_26	Leveraged / Inverse	0.079	0.98	0.9

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 3 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_16	Real Estate	2.923	1.000	1.00	1.000	0.99
cluster_5	U.S. Large Cap Equity	0.846	0.952	0.56	0.869	0.32
cluster_1	U.S. Mid Cap Equity	2.691	1.000	1.00	0.279	0.22

Imminent SELL (top 2 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_4	Developed ex-US Equity	0.012	0.527	0.58	-0.487	-0.51
cluster_25	International Fixed Income	0.091	0.146	0.21	-0.146	-0.21

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 141.739 20dma: 142.001 50dma: 141.438

Spread vs 50dma: +0.301 Spread vs 20dma: -0.262

P(buy crossover within 5 days): 26.6% P(sell): 83.3%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
USMV	cluster_7	U.S. Low Volatility Equity	0.983	26	2.1%	11.66%	0.93
XLU	cluster_7	U.S. Low Volatility Equity	0.983	26	2.1%	4.39%	0.93
VTV	cluster_2	U.S. Large Cap Value	0.939	67	6.7%	22.37%	0.97
BBAX	cluster_9	Asia-Pacific Equity	0.933	2	-0.3%	7.51%	0.97
IGF	cluster_9	Asia-Pacific Equity	0.933	2	-0.3%	4.54%	0.97
EWZ	cluster_9	Asia-Pacific Equity	0.933	2	-0.3%	1.46%	0.97
PDBC	cluster_19	Broad Commodities	0.653	31	-5.2%	8.73%	0.98
VHT	cluster_13	Healthcare	0.434	50	9.6%	12.84%	0.93
BWX	cluster_25	International Fixed Income	-0.212	34	-2.4%	15.01%	0.93
VEA	cluster_4	Developed ex-US Equity	-0.506	10	-2.5%	11.57%	0.99

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
IJH	cluster_1	U.S. Mid Cap Equity	1	0.215	1.007	0.99
VNQ	cluster_16	Real Estate	1	0.990	0.971	0.99

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
16	cluster_16	Real Estate	Real Estate	14	FLAT				0.990	+0.024
7	cluster_7	U.S. Low Volatility Equity	U.S. Low Volatili...	30	LONG		26	2.1%	0.983	-0.006
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	65	LONG		67	6.7%	0.939	-0.048
9	cluster_9	Asia-Pacific Equity	Asia-Pacific Equi...	24	LONG		2	-0.3%	0.933	+0.082
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				0.900	+0.126
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		31	-5.2%	0.653	+0.257
13	cluster_13	Healthcare	Healthcare	17	LONG		50	9.6%	0.434	-0.214
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	51	FLAT				0.319	-0.268
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	83	FLAT				0.215	-0.013
14	cluster_14	China Equity	China Equity	15	FLAT				0.001	+0.064
25	cluster_25	International Fixed Income	International Fix...	6	LONG		34	-2.4%	-0.212	+0.198
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				-0.234	+0.253
20	cluster_20	India Equity	India Equity	10	FLAT				-0.413	+0.078
10	cluster_10	Materials	Materials	21	FLAT				-0.417	+0.059
4	cluster_4	Developed ex-US Equity	Developed ex-US E...	61	LONG		10	-2.5%	-0.506	+0.020
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.782	+0.152
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				-0.931	-0.204
18	cluster_18	Gold	Gold	12	FLAT				-0.939	-0.002
6	cluster_6	Aggregate Fixed Income	Aggregate Fixed I...	43	FLAT				-0.954	+0.036
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.960	+0.025
3	cluster_3	U.S. Large Cap Growth	U.S. Large Cap Gr...	64	FLAT				-0.961	-0.038
11	cluster_11	High Yield Bond	High Yield Bond	21	FLAT				-0.993	+0.006
17	cluster_17	Municipal Bond	Municipal Bond	13	FLAT				-0.995	-0.041
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	FLAT				-0.999	-0.000
12	cluster_12	U.S. Momentum Equity	U.S. Momentum Equ...	20	FLAT				-1.000	-0.000
15	cluster_15	Fixed Income	Fixed Income	15	FLAT				-1.000	+0.000

Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 83 **Var explained:** 0.872 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +28.8%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +20.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1265
Consumer Discretionary	11	0.1210
U.S. Mid Cap Equity	6	0.0761
U.S. Small Cap Value	6	0.0753
U.S. Small Cap Growth	6	0.0741
Industrials	5	0.0552
U.S. Mid Cap Value	3	0.0380
U.S. Mid Cap Growth	3	0.0371

Utilities	3	0.0334
Leveraged / Inverse	2	0.0242
Dividend Equity	2	0.0241
U.S. Small Cap Low Volatility	2	0.0230
U.S. Small Cap Fundamental Equity	1	0.0128
U.S. Small Cap ESG Equity	1	0.0128
Unknown	1	0.0128
U.S. Mid/Small Cap Equity	1	0.0127
Mid Cap Value Equities	1	0.0127
U.S. Small/Mid Cap Fundamental Equity	1	0.0127
U.S. Small Cap Multifactor Equity	1	0.0126
Small Cap Value Equities	1	0.0125
U.S. Small/Mid Cap Equity	1	0.0124
U.S. ESG Equity	1	0.0124
U.S. Extended Market Equity	1	0.0124
U.S. Mid Cap Dividend Equity	1	0.0124
U.S. Mid Cap Quality Equity	1	0.0123
Growth Equity	1	0.0123
Value Equity	1	0.0122
U.S. Equity	1	0.0122
U.S. Small Cap Dividend Equity	1	0.0122
U.S. Small Cap Industrials	1	0.0120
U.S. Small Cap Momentum Equity	1	0.0120
U.S. Micro Cap Equity	1	0.0117
U.S. Small Cap Consumer Discretionary	1	0.0116
U.S. Dividend Equity	1	0.0114
U.S. Mid Cap Low Volatility	1	0.0110
Real Estate	1	0.0099

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
IJH	1.007	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
18	NUSC	0.0128	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
22	MDY	0.0128	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
30	IVOO	0.0128	Vanguard S&P Mid-Cap 400 ETF		FLAT
37	IJH	0.0128	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

38	SPMD	0.0128	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
40	VB	0.0128	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
56	FNDA	0.0128	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
15	FNX	0.0127	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
19	PRFZ	0.0127	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
23	IJT	0.0127	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
31	VI00	0.0127	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
34	IJR	0.0127	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
39	SPSM	0.0127	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
41	SCHA	0.0127	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
52	EZM	0.0127	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
63	IVOV	0.0127	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
64	MDYV	0.0127	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
68	VBR	0.0127	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
5	IWM	0.0126	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
12	JHMM	0.0126	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
14	SCHM	0.0126	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	FLAT
16	FYX	0.0126	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
20	SMLF	0.0126	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
25	VIOG	0.0126	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
28	SLYG	0.0126	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
32	VTWO	0.0126	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
59	IWN	0.0126	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
61	IJJ	0.0126	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
75	RWK	0.0126	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
51	EES	0.0125	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
53	PSC	0.0125	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
62	IJS	0.0125	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
65	SLYV	0.0125	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
66	VIOV	0.0125	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
67	VTWV	0.0125	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
82	SDVY	0.0125	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
24	IJK	0.0124	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	FLAT
26	IVOG	0.0124	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	FLAT
27	MDYG	0.0124	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	FLAT
36	VXF	0.0124	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
50	DON	0.0124	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
69	ETHO	0.0124	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
70	RWJ	0.0124	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
83	ISMD	0.0124	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
35	ISCG	0.0123	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
72	XMHQ	0.0123	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
49	DES	0.0122	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
74	RFV	0.0122	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
76	OMFS	0.0122	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT

81	FYT	0.0122	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
21	IWO	0.0121	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	FLAT
29	VTWG	0.0121	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	FLAT
42	FXD	0.0121	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
33	VBK	0.0120	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
71	PSCI	0.0120	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
77	RZV	0.0120	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
80	XSMO	0.0120	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
47	RSPD	0.0119	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
17	IWC	0.0117	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
60	SMMV	0.0117	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
11	FIW	0.0116	First Trust Water ETF	Utilities	LONG
13	REGL	0.0116	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
44	FTXR	0.0116	First Trust Nasdaq Transportation ETF	Industrials	LONG
79	PSCD	0.0116	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
48	SMDV	0.0114	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
9	PHO	0.0113	Invesco Water Resources ETF	Utilities	LONG
55	XTN	0.0113	State Street SPDR S&P Transportation ETF	Industrials	FLAT
58	XSLV	0.0113	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
73	PKB	0.0113	Invesco Building & Construction ETF	Industrials	FLAT
2	XRT	0.0112	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
6	IYC	0.0112	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
7	VCR	0.0111	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
8	FDIS	0.0111	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
46	IYT	0.0110	iShares U.S. Transportation ETF	Industrials	FLAT
57	XMLV	0.0110	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
43	PEJ	0.0109	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
54	XHB	0.0109	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	FLAT
1	XLY	0.0107	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	CGW	0.0106	Invesco S&P Global Water Index ETF	Utilities	LONG
3	JETS	0.0100	U.S. Global Jets ETF	Industrials	FLAT
45	ITB	0.0100	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT
4	REM	0.0099	iShares Mortgage Real Estate ETF	Real Estate	LONG
78	RTH	0.0099	VanEck Retail ETF	Consumer Discretionary	LONG

Cluster 2 – U.S. Large Cap Value

Members: 65 **Var explained:** 0.881 **Signal:** LONG since 2026-04-13 (67 days, 6.7%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.7%)

- Latest sell_cross: 2026-03-05
- Current 252d return: +18.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-05)



Category breakdown

category	n_members	abs_weight
Financials	13	0.1869
U.S. Large Cap Value	10	0.1607
U.S. Dividend Equity	7	0.1096
Industrials	6	0.0913
Value Equity	4	0.0620
U.S. Mid Cap Equity	3	0.0474
U.S. Multifactor Equity	3	0.0472
U.S. Value Factor Equity	3	0.0465
U.S. Fundamental Equity	2	0.0324
U.S. Mid Cap Value	2	0.0321
U.S. Equal Weight Equity	2	0.0321
U.S. Equity	2	0.0316
Large Cap Blend Equities	1	0.0160
U.S. Revenue Weighted Equity	1	0.0159
U.S. Size Factor Equity	1	0.0158
U.S. Buyback Equity	1	0.0158
U.S. Mega Cap Value	1	0.0157
U.S. Moat Equity	1	0.0148

Global Equity	1	0.0135
Communication Services	1	0.0127

Replication basket – VTV (cor=0.97) basket lead at lag 1: 0.003

ticker	rep_wt	name	category	etf_signal
VTV	1.127	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
6	RSP	0.0162	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
14	PRF	0.0162	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
21	FNDX	0.0162	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
25	IWD	0.0162	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
34	VONV	0.0162	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
17	CFA	0.0161	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
38	ILCV	0.0161	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
40	IUSV	0.0161	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
41	SCHV	0.0161	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	FLAT
52	IWS	0.0161	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
9	FEX	0.0160	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
27	IVE	0.0160	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
28	JPUS	0.0160	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
32	VOOV	0.0160	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
42	SPYV	0.0160	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
44	VTV	0.0160	Vanguard Value ETF	U.S. Large Cap Value	LONG
56	VOE	0.0160	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
15	RWL	0.0159	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
19	NULV	0.0159	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
22	EQAL	0.0159	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
23	IWX	0.0159	iShares Russell Top 200 Value ETF	Value Equity	LONG
37	VYM	0.0159	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
39	IMCB	0.0159	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
58	DTD	0.0159	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
62	JPME	0.0159	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
8	PKW	0.0158	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
26	IWR	0.0158	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
29	SIZE	0.0158	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
63	JVAL	0.0158	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	FLAT
11	RDVY	0.0157	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT

16	AIVL	0.0157	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
24	ONEV	0.0157	State Street SPDR Russell 1000 Low Volatility	U.S. Equity	FLAT
35	MGV	0.0157	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
43	VO	0.0157	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
46	FTA	0.0157	First Trust Large Cap Value AlphaDEX Fund	Value Equity	LONG
57	DLN	0.0157	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
13	IYJ	0.0156	iShares U.S. Industrials ETF	Industrials	FLAT
20	FDIV	0.0156	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
53	ONEY	0.0156	State Street SPDR Russell 1000 Yield Focus	U.S. Dividend Equity	FLAT
7	FXR	0.0153	First Trust Industrials AlphaDEX	Industrials	FLAT
10	PWV	0.0153	Invesco Large Cap Value ETF	Value Equity	LONG
48	IYF	0.0153	iShares U.S. Financial ETF	Financials	FLAT
54	VFH	0.0153	Vanguard Financials ETF	Financials	FLAT
55	FNCL	0.0153	Fidelity MSCI Financials Index ETF	Financials	LONG
18	CFO	0.0152	VictoryShares US 500 Enhanced Volatility	U.S. Multifactor Equity	FLAT
31	VIS	0.0152	Vanguard Industrials ETF	Industrials	FLAT
33	FIDU	0.0152	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
49	IYG	0.0152	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0151	State Street Financial Select Sector SPDR	Financials	LONG
2	XLI	0.0151	State Street Industrial Select Sector SPDR	Industrials	FLAT
36	SCHD	0.0151	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
45	FXO	0.0151	First Trust Financials AlphaDEX	Financials	LONG
51	RPV	0.0151	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
30	VLUE	0.0150	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	FLAT
12	MOAT	0.0148	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
59	PAVE	0.0148	Global X U.S. Infrastructure Development	Industrials	FLAT
65	KCE	0.0148	State Street SPDR S&P Capital Markets	Financials	LONG
50	KBWB	0.0141	Invesco KBW Bank ETF	Financials	FLAT
64	IAI	0.0140	iShares U.S. Broker-Dealers & Securities	Financials	LONG
4	KBE	0.0136	State Street SPDR S&P Bank ETF	Financials	FLAT
61	PSP	0.0135	Invesco Global Listed Private Equity ETF	Global Equity	FLAT
47	IAT	0.0133	iShares U.S. Regional Banks ETF	Financials	LONG
3	KRE	0.0130	State Street SPDR S&P Regional Banking	Financials	FLAT
5	KIE	0.0130	State Street SPDR S&P Insurance ETF	Financials	LONG
60	IYZ	0.0127	iShares U.S. Telecommunications ETF	Communication Services	FLAT

Cluster 3 – U.S. Large Cap Growth**Members:** 64 **Var explained:** 0.821 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +33.9%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +20.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-17 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Technology	34	0.5283
U.S. Large Cap Growth	11	0.1851
Communication Services	3	0.0439
U.S. Equity	2	0.0333
Growth Equity	2	0.0330
Financials	2	0.0292
Consumer Discretionary	2	0.0283
U.S. Mega Cap Growth	1	0.0167
U.S. Equal Weight Equity	1	0.0166
IPO Equity	1	0.0152
Options Income	1	0.0151
U.S. Small Cap Technology	1	0.0151
Trend-Following Equity	1	0.0147
U.S. Trend-Following Equity	1	0.0129
Israel Equity	1	0.0126

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
QQQ	1.026	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
16	QQQ	0.0170	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
37	IGM	0.0170	iShares Expanded Tech Sector ETF	Technology	FLAT
41	ONEQ	0.0170	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
52	ILCG	0.0170	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
56	VUG	0.0169	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
31	XNTK	0.0168	State Street SPDR NYSE Technology ETF	Technology	FLAT
39	IYW	0.0168	iShares U.S. Technology ETF	Technology	FLAT
40	NULG	0.0168	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
43	IWF	0.0168	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
45	VGT	0.0168	Vanguard Information Tech ETF	Technology	FLAT
49	FTEC	0.0168	Fidelity MSCI Information Technology In...	Technology	FLAT
50	VONG	0.0168	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
53	IUSG	0.0168	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
54	SCHG	0.0168	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
20	QTEC	0.0167	First Trust NASDAQ-100-Technology Secto...	Technology	FLAT
26	FXL	0.0167	First Trust Technology AlphaDEX	Technology	FLAT
44	IVW	0.0167	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
46	VOOG	0.0167	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
51	MGK	0.0167	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
55	SPYG	0.0167	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0166	State Street Technology Select Sector S...	Technology	FLAT
9	AIQ	0.0166	Global X Artificial Intelligence & Tech...	Technology	FLAT
19	QQEW	0.0166	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	FLAT
42	IWY	0.0166	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
23	XT	0.0165	iShares Future Exponential Technologies...	Technology	FLAT
38	IXN	0.0165	iShares Global Tech ETF	Technology	FLAT
36	PWB	0.0163	Invesco Large Cap Growth ETF	Growth Equity	FLAT
57	SPHB	0.0163	Invesco S&P 500 High Beta ETF	U.S. Equity	FLAT
7	BOTZ	0.0158	Global X Robotics & Artificial Intellig...	Technology	FLAT
8	ROBO	0.0158	ROBO Global Robotics and Automation Ind...	Technology	FLAT
22	TDIV	0.0158	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
30	FDN	0.0158	First Trust DJ Internet Index Fund	Technology	LONG
64	QTUM	0.0157	Defiance Quantum ETF	Technology	LONG
4	SMH	0.0155	VanEck Semiconductor ETF	Technology	FLAT

29	PNQI	0.0155	Invesco Nasdaq Internet ETF	Technology	FLAT
6	SKYY	0.0154	First Trust Cloud Computing ETF	Technology	LONG
3	SOXX	0.0153	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
14	ARKQ	0.0153	ARK Autonomous Technology & Robotics ETF	Technology	FLAT
17	NXTG	0.0152	First Trust Indxx NextG ETF	Technology	FLAT
24	XSD	0.0152	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
28	IPO	0.0152	Renaissance IPO ETF	IPO Equity	FLAT
35	QYLD	0.0151	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
60	PSCT	0.0151	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
63	PTF	0.0151	Invesco Dorsey Wright Technology Moment...	Technology	LONG
5	IGV	0.0150	iShares Expanded Tech-Software Sector E...	Technology	FLAT
13	ARKW	0.0150	ARK Next Generation Internet ETF	Technology	LONG
33	FINX	0.0150	Global X FinTech ETF	Financials	FLAT
21	PSI	0.0149	Invesco Semiconductors ETF	Technology	FLAT
62	FTXL	0.0149	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
12	ARKK	0.0148	ARK Innovation ETF	Technology	FLAT
47	VOX	0.0148	Vanguard Communication Services ETF	Communication Services	LONG
48	FCOM	0.0148	Fidelity MSCI Communication Services In...	Communication Services	LONG
34	PTNQ	0.0147	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
25	IBUY	0.0145	Amplify Online Retail ETF	Consumer Discretionary	LONG
27	CIBR	0.0144	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
18	PRNT	0.0143	3D Printing (The) ETF	Technology	FLAT
2	XLC	0.0142	State Street Communication Services Sel...	Communication Services	FLAT
11	HACK	0.0142	Amplify Cybersecurity ETF	Technology	FLAT
32	IPAY	0.0142	Amplify Digital Payments ETF	Financials	FLAT
15	ESPO	0.0138	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
10	BLOK	0.0137	Amplify Blockchain Technology ETF	Technology	FLAT
58	PTLC	0.0129	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT
59	EIS	0.0126	iShares MSCI Israel ETF	Israel Equity	FLAT
61	BLCN	0.0120	Siren NexGen Economy ETF	Technology	FLAT

Cluster 4 – Developed ex-US Equity

Members: 61 **Var explained:** 0.894 **Signal:** LONG since 2026-07-06 (10 days, -2.5%)

Signal logic:

- Latest buy_cross: 2026-07-06 – filter **PASS** (252d at date = +21.5%)
- Latest sell_cross: 2026-06-25
- Current 252d return: +18.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-06 > last sell at 2026-06-25)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3158
Developed ex-US Equity	9	0.1528
Global ex-US Equity	5	0.0848
Europe Equity	4	0.0675
International Dividend Equity	4	0.0659
Alternatives	4	0.0615
Eurozone Equity	3	0.0490
Commodity	1	0.0172
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162
Italy Equity	1	0.0159
Europe Financials	1	0.0158
United Kingdom Equity	1	0.0158
Netherlands Equity	1	0.0156
Sweden Equity	1	0.0156
Spain Equity	1	0.0151
Austria Equity	1	0.0148
Belgium Equity	1	0.0145
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
VEA	0.96	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0173	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
51	IEFA	0.0173	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
40	GSIE	0.0172	Goldman Sachs ActiveBeta International ...	Commodity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	FLAT
32	INTF	0.0171	iShares International Equity Factor ETF	International Equity	LONG
6	IEV	0.0170	iShares Europe ETF	Europe Equity	LONG
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	FLAT
19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	LONG
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
49	VGK	0.0170	Vanguard FTSEEurope ETF	Europe Equity	LONG
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	FLAT
14	EZU	0.0168	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
26	JPIN	0.0168	JPMorgan Diversified Return Internation...	International Equity	FLAT
38	VYMI	0.0168	Vanguard International High Dividend Yi...	International Dividend E...	LONG
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	LONG
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	FLAT
34	IVLU	0.0167	iShares MSCI Intl Value Factor ETF	International Equity	LONG
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	FLAT
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT

36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	FLAT
31	IMTM	0.0164	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0162	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	LONG
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	FLAT
13	EWU	0.0158	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
15	IDV	0.0158	iShares International Select Dividend E...	International Dividend E...	FLAT
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	LONG
27	DBEF	0.0157	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
42	HDEF	0.0156	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	FLAT
58	EWN	0.0156	iShares MSCI Netherlands Index Fund	Netherlands Equity	FLAT
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	FLAT
11	EWSP	0.0151	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWAO	0.0148	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWBK	0.0145	iShares MSCI Belgium ETF	Belgium Equity	FLAT
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	LONG

Cluster 5 – U.S. Large Cap Equity

Members: 51 **Var explained:** 0.941 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.2%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +17.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	10	0.2000
U.S. Total Market Equity	7	0.1410
Global Equity	7	0.1381
U.S. Dividend Equity	4	0.0761
Asset Allocation	4	0.0731
U.S. Mega Cap Equity	3	0.0591
U.S. Quality / Factor Equity	3	0.0588
U.S. Multifactor Equity	3	0.0587
U.S. ESG Equity	2	0.0402
U.S. Equity	2	0.0401
Dividend Equity	2	0.0378
Commodity	1	0.0201
Large Cap Value Equities	1	0.0200
Large Cap Blend Equities	1	0.0199
Alternatives	1	0.0171

Replication basket – VOO (cor=1.00) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
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V00 0.918 Vanguard S&P 500 ETF U.S. Large Cap Equity LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
30	IWB	0.0202	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
48	SPTM	0.0202	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
1	SPY	0.0201	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
18	JHML	0.0201	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
22	IWV	0.0201	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
23	IYY	0.0201	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
26	SPYX	0.0201	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
29	ESGU	0.0201	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
35	VTHR	0.0201	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
36	GSLC	0.0201	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
39	VONE	0.0201	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
42	VV	0.0201	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
43	ILCB	0.0201	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
44	ITOT	0.0201	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
45	IVV	0.0201	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
46	SCHB	0.0201	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
47	SCHX	0.0201	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
49	VOO	0.0201	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
50	VTI	0.0201	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
8	URTH	0.0200	iShares MSCI World ETF	Global Equity	LONG
21	SUSA	0.0200	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
24	LRGF	0.0200	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
31	IWL	0.0200	iShares Russell Top 200 ETF	U.S. Equity	LONG
38	EPS	0.0200	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
40	MGC	0.0200	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
3	ACWI	0.0199	iShares MSCI ACWI ETF	Global Equity	FLAT
12	TILT	0.0199	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
20	DSI	0.0199	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
9	CRBN	0.0198	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
11	VT	0.0198	Vanguard Total World Stock Index ETF	Global Equity	FLAT
32	QUAL	0.0198	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
4	AOA	0.0197	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
10	SPGM	0.0197	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
16	QDF	0.0197	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
25	OEF	0.0197	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
33	QUS	0.0197	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
17	CATH	0.0195	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
19	DGRW	0.0195	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG

15	I00	0.0194	iShares Global 100 ETF	Global Equity	LONG
27	XLG	0.0193	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
34	SPHQ	0.0193	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	FLAT
41	VIG	0.0192	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
7	AOR	0.0191	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
13	PFM	0.0189	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
28	DIA	0.0188	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
51	OMFL	0.0188	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
37	DGRO	0.0185	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
14	OUSA	0.0181	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
6	AOM	0.0179	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
2	QAI	0.0171	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
5	AOK	0.0164	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT

Cluster 6 – Aggregate Fixed Income

Members: 43 **Var explained:** 0.842 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-25 – filter **PASS** (252d at date = +3.8%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +3.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-25)



Category breakdown

category	n_members	abs_weight
Fixed Income	13	0.3082
Aggregate Fixed Income	6	0.1478
Investment Grade Corporate Bond	6	0.1441
Mortgage-Backed Fixed Income	6	0.1326
Active Fixed Income	4	0.0943
Cash / Ultra-Short Fixed Income	3	0.0668
Target Maturity Bond	2	0.0419
International Fixed Income	2	0.0417
Agency Fixed Income	1	0.0227

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.009

ticker	rep_wt	name	category	etf_signal
BND	0.729	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

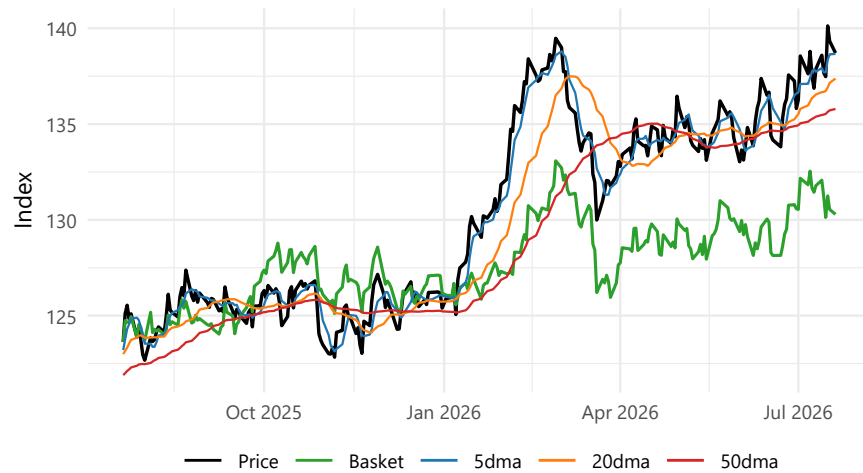
	ticker	PC_wt	name	category	etf_signal
13	BIV	0.0250	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
16	AGG	0.0249	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
17	BND	0.0249	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
33	VGIT	0.0249	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
19	SCHZ	0.0248	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
29	IEI	0.0248	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
30	SPTI	0.0248	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
32	SCHR	0.0248	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
6	GVI	0.0247	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
12	IUSB	0.0247	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
20	SPAB	0.0247	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
1	IEF	0.0246	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
11	ISTB	0.0245	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
9	SPIB	0.0243	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
10	IGIB	0.0243	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
14	BSV	0.0242	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
15	VCIT	0.0242	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
35	MBB	0.0242	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
36	VMBS	0.0242	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
5	FBND	0.0241	Fidelity Total Bond ETF	Active Fixed Income	FLAT
37	SPMB	0.0241	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
24	IGSB	0.0240	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
26	VCSH	0.0239	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
3	BOND	0.0238	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
4	GTO	0.0238	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
7	AGGY	0.0238	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
25	SLQD	0.0234	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
31	GOVT	0.0233	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
34	GNMA	0.0233	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
38	SHY	0.0229	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
28	AGZ	0.0227	iShares Agency Bond ETF	Agency Fixed Income	FLAT
18	TOTL	0.0226	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
23	SPSB	0.0226	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
41	VGSH	0.0226	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
39	SPTS	0.0221	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
40	SCHO	0.0221	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
22	BSCQ	0.0212	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	LONG
42	BNDX	0.0210	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
8	IBDR	0.0207	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
43	IAGG	0.0207	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
27	CMBS	0.0186	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
2	LMBS	0.0181	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
21	LDUR	0.0172	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 7 – U.S. Low Volatility Equity

Members: 30 **Var explained:** 0.777 **Signal:** LONG since 2026-06-10 (26 days, 2.1%)

Signal logic:

- Latest buy_cross: 2026-06-10 – filter **PASS** (252d at date = +11.9%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +12.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-10 > last sell at 2026-06-03)

**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	11	0.3828
Consumer Staples	6	0.1908
U.S. Low Volatility Equity	4	0.1417
Utilities	4	0.1183
Dividend Equity	1	0.0362
U.S. Defensive Equity	1	0.0352
U.S. Quality / Factor Equity	1	0.0342
Global Equity	1	0.0331
U.S. Small Cap Consumer Staples	1	0.0279

Replication basket – XLU+USMV (cor=0.93) basket lead at lag 1: 0.013

ticker	rep_wt	name	category	etf_signal
XLU	0.253	State Street Utilities Select Sector ...	Utilities	LONG
USMV	0.672	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility E...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	FVD	0.0369	First Trust VL Dividend	U.S. Dividend Equity	LONG
9	LVHD	0.0366	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
8	SDY	0.0365	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
16	SPHD	0.0362	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
14	DVY	0.0357	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
21	SPLV	0.0356	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
23	LGLV	0.0356	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
6	DHS	0.0352	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
7	CDC	0.0352	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
17	SPYD	0.0351	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
10	HDV	0.0347	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
13	SDOG	0.0346	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
5	FDL	0.0343	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
18	FTCS	0.0342	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
12	DIV	0.0339	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
22	USMV	0.0338	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
11	PEY	0.0333	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
3	ACWV	0.0331	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
15	RDIV	0.0327	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
24	VDC	0.0324	Vanguard Consumer Staples ETF	Consumer Staples	LONG
25	FSTA	0.0324	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0318	State Street Consumer Staples Select Se...	Consumer Staples	LONG
30	PBJ	0.0316	Invesco Food & Beverage ETF	Consumer Staples	LONG
19	KXI	0.0315	iShares Global Consumer Staples ETF	Consumer Staples	LONG
20	IYK	0.0312	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
26	IDU	0.0300	iShares U.S. Utilities ETF	Utilities	LONG
27	VPU	0.0296	Vanguard Utilities ETF	Utilities	LONG
28	FUTY	0.0296	Fidelity MSCI Utilities Index ETF	Utilities	LONG
2	XLU	0.0291	State Street Utilities Select Sector SP...	Utilities	LONG
29	PSCC	0.0279	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 8 – Emerging Markets Equity**Members:** 25 **Var explained:** 0.872 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +24.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-04-10)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7728
Asia Equity	2	0.0814
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0362
South Korea Equity	1	0.0341
Global Dividend Equity	1	0.0337

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.035

ticker	rep_wt	name	category	etf_signal
IEMG	0.92	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
11	EEMA	0.0414	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0409	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
10	AIA	0.0403	iShares Asia 50 ETF	Asia Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	FLAT
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0388	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0364	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWI	0.0362	iShares MSCI Taiwan ETF	Taiwan Equity	FLAT
25	FEMS	0.0349	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0337	Global X SuperDividend ETF	Global Dividend Equity	FLAT

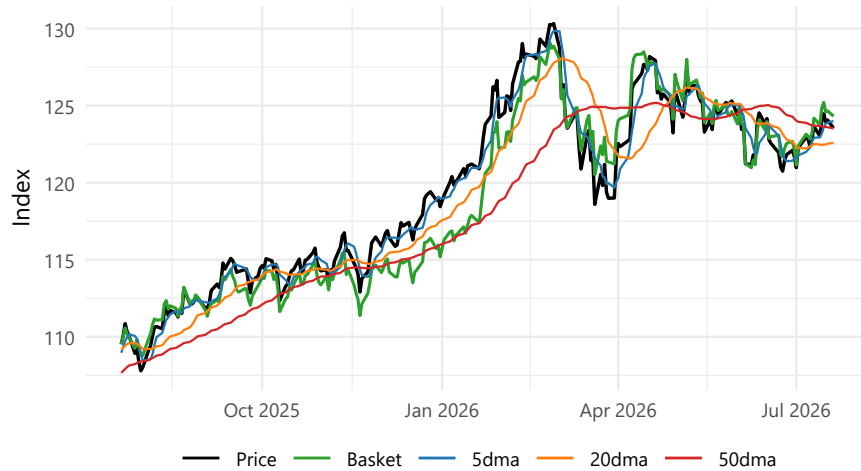
Cluster 9 – Asia-Pacific Equity

Members: 24 **Var explained:** 0.680 **Signal:** LONG since 2026-07-16 (2 days, -0.3%)

Signal logic:

- Latest buy_cross: 2026-07-16 – filter **PASS** (252d at date = +14.5%)

- Latest sell_cross: 2026-06-23
- Current 252d return: +13.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-16 > last sell at 2026-06-23)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0944
International Equity	2	0.0919
International Dividend Equity	2	0.0919
Global Equity	2	0.0881
Real Estate	2	0.0881
Emerging Markets Fixed Income	2	0.0803
Currency	2	0.0778
Australia Equity	1	0.0467
Canada Equity	1	0.0444
Switzerland Equity	1	0.0413
Singapore Equity	1	0.0399
South Africa Equity	1	0.0395
Latin America Equity	1	0.0379
New Zealand Equity	1	0.0379
Mexico Equity	1	0.0359
Brazil Equity	1	0.0320
Chile Equity	1	0.0319

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
EWZ	0.091	iShares MSCI Brazil ETF	Brazil Equity	LONG
IGF	0.281	iShares Global Infrastructure ETF	Global Equity	LONG
BBAX	0.465	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	EPP	0.0472	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
24	BBAX	0.0472	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
12	IDLV	0.0469	Invesco S&P International Developed Low...	International Equity	LONG
4	EWA	0.0467	iShares MSCI Australia Index Fund	Australia Equity	LONG
11	DWX	0.0463	State Street SPDR S&P International Div...	International Dividend E...	LONG
3	PID	0.0456	Invesco International Dividend Achiever...	International Dividend E...	LONG
13	EFAV	0.0450	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
9	NFRA	0.0446	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
14	VNQI	0.0446	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
7	EWC	0.0444	iShares MSCI Canada Index Fund	Canada Equity	LONG
2	RWX	0.0435	State Street SPDR Dow Jones Internation...	Real Estate	LONG
10	IGF	0.0435	iShares Global Infrastructure ETF	Global Equity	LONG
20	FXA	0.0414	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
5	EWL	0.0413	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
15	EBND	0.0403	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
16	EMLC	0.0400	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
6	EWS	0.0399	iShares MSCI Singapore ETF	Singapore Equity	LONG
22	EZA	0.0395	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
19	ILF	0.0379	iShares Latin America 40 ETF	Latin America Equity	FLAT
21	ENZL	0.0379	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
23	FXC	0.0364	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
17	EWX	0.0359	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0320	iShares MSCI Brazil ETF	Brazil Equity	LONG
18	ECH	0.0319	iShares MSCI Chile ETF	Chile Equity	FLAT

Cluster 10 – Materials

Members: 21 **Var explained:** 0.699 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.4%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +24.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6550
MLP / Energy Infrastructure	4	0.1680
Energy	1	0.0502
Consumer Staples	1	0.0493
Norway Equity	1	0.0438
Industrial Metals	1	0.0337

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.043

ticker	rep_wt	name	category	etf_signal
AMLP	0.106	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.084	Global X Uranium ETF	Materials	FLAT
GUNR	0.776	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	GNR	0.0551	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0544	FlexShares Global Upstream Natural Reso...	Materials	FLAT
11	IYM	0.0530	iShares U.S. Basic Materials ETF	Materials	FLAT
10	MXI	0.0529	iShares Global Materials ETF	Materials	FLAT
21	NANR	0.0526	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0514	Vanguard Materials ETF	Materials	FLAT
16	PICK	0.0509	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0506	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
1	XLB	0.0505	State Street Materials Select Sector SP...	Materials	FLAT
20	IGE	0.0502	iShares North American Natural Resource...	Energy	FLAT
4	XME	0.0499	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
17	SLX	0.0498	VanEck Steel ETF	Materials	FLAT
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	LONG
3	COPX	0.0472	Global X Copper Miners ETF	Materials	FLAT
18	NORW	0.0438	Global X MSCI Norway ETF	Norway Equity	FLAT
5	EMLP	0.0435	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
9	MLPX	0.0434	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
7	AMLP	0.0411	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
8	MLPA	0.0400	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0367	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0337	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 11 – High Yield Bond

Members: 21 **Var explained:** 0.765 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +4.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5157
Preferred Stock	7	0.3058
Emerging Markets Fixed Income	4	0.1785

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0517	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0460	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 12 – U.S. Momentum Equity

Members: 20 **Var explained:** 0.822 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +44.7%)
- Latest sell_cross: 2026-07-08
- Current 252d return: +17.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-08 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Industrials	4	0.1845
U.S. Mid Cap Growth	3	0.1567
U.S. Momentum Equity	3	0.1534
Momentum Equity	2	0.1033
Convertible Bond	2	0.0996
Technology	1	0.0533
Growth Equity	1	0.0521
IPO Equity	1	0.0516
U.S. Mid Cap Momentum Equity	1	0.0515
U.S. Small Cap Momentum Equity	1	0.0503
International IPO Equity	1	0.0436

Replication basket – MTUM (cor=0.91) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
MTUM	0.836	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	FTC	0.0533	First Trust Large Cap Growth AlphaDEX F...	Technology	FLAT
4	PDP	0.0533	Invesco Dorsey Wright Momentum ETF	Momentum Equity	FLAT
19	VFMO	0.0532	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	FLAT
13	IMCG	0.0526	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
7	RPG	0.0521	Invesco S&P 500 Pure Growth ETF	Growth Equity	FLAT
12	VOT	0.0521	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
9	IWP	0.0520	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	FLAT
5	FPX	0.0516	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
8	XMMO	0.0515	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	FLAT
6	CWB	0.0503	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
11	MTUM	0.0503	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	FLAT
14	DWAS	0.0503	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
20	PRN	0.0502	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
1	FV	0.0500	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	FLAT
18	QMOM	0.0499	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
10	ICVT	0.0492	iShares Convertible Bond ETF	Convertible Bond	FLAT
17	XAR	0.0462	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
15	PPA	0.0452	Invesco Aerospace & Defense ETF	Industrials	FLAT
2	FPXI	0.0436	First Trust International Equity Opport...	International IPO Equity	FLAT
16	ITA	0.0429	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT

Cluster 13 – Healthcare

Members: 17 **Var explained:** 0.758 **Signal:** LONG since 2026-05-06 (50 days, 9.6%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +26.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VHT	1.092	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0655	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0654	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0647	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0641	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0639	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0624	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0622	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0568	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0545	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0538	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0532	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0506	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0472	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 14 – China Equity

Members: 15 **Var explained:** 0.797 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +1.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8765
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0592

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.010

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0739	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0737	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0731	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0711	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0596	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
12	EWK	0.0592	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
14	CNXT	0.0591	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0582	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 15 – Fixed Income

Members: 15 **Var explained:** 0.928 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-15 – filter **PASS** (252d at date = +3.9%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +2.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-15)



Category breakdown

category	n_members	abs_weight
Investment Grade Corporate Bond	7	0.4707
Fixed Income	6	0.4009
Active Fixed Income	1	0.0665
Taxable Municipal Bond	1	0.0619

Replication basket – TLT (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
TLT	0.694	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
9	BLV	0.0687	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
7	SPLB	0.0678	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
8	IGLB	0.0678	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
10	VCLT	0.0676	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
2	LQD	0.0675	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
11	USIG	0.0672	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
6	QLTA	0.0671	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
12	TLH	0.0668	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
14	SPTL	0.0667	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
15	VGLT	0.0666	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
3	FLCO	0.0665	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
1	TLT	0.0664	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
5	CORP	0.0657	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
13	EDV	0.0656	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
4	BAB	0.0619	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT

Cluster 16 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +16.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT

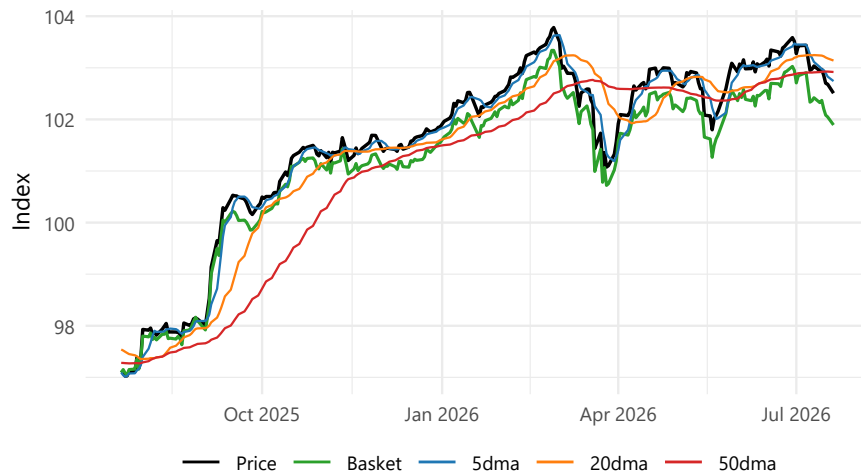
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0697	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 17 – Municipal Bond

Members: 13 Var explained: 0.819 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-07-10
- Current 252d return: +5.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-10 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0802	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0794	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0794	iShares New York Muni Bond ETF	Municipal Bond	FLAT
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0747	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0721	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	FLAT
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 18 – Gold

Members: 12 **Var explained:** 0.862 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +21.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1724
Silver	2	0.1617

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
GLD	0.014	SPDR Gold Shares	Gold	FLAT
IAU	0.757	iShares Gold Trust Shares	Gold	FLAT
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT

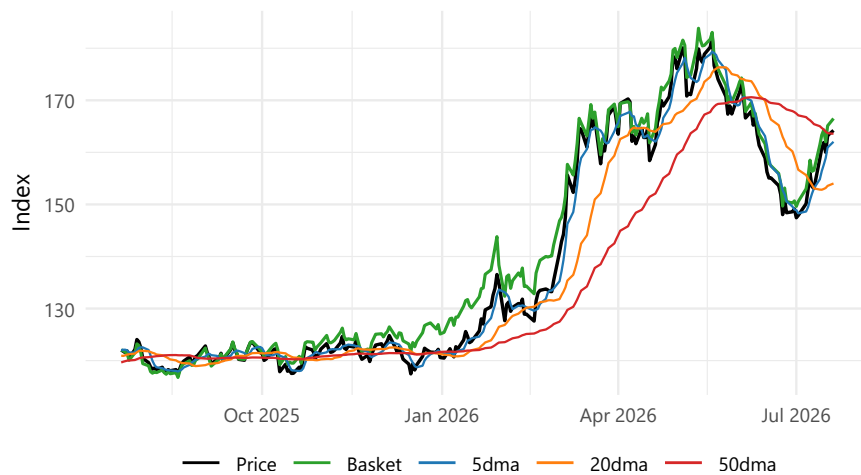
2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0810	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.771 **Signal:** LONG since 2026-06-03 (31 days, -5.2%)

Signal logic:

- Latest buy_cross: 2026-06-03 – filter **PASS** (252d at date = +48.1%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +34.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-03 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6174
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0498
Natural Gas	1	0.0328

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.076

ticker	rep_wt	name	category	etf_signal
PDBC	1.122	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1056	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
6	GSG	0.1056	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1056	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1052	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.0995	United States Oil Fund	Oil	LONG
5	DBO	0.0985	Invesco DB Oil Fund	Oil	FLAT
7	DJP	0.0983	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.0972	First Trust Global Tactical Commodity S...	Broad Commodities	LONG
3	DBA	0.0498	Invesco DB Agriculture Fund	Agricultural Commodities	LONG
11	UNG	0.0328	United States Natural Gas Fund LP	Natural Gas	FLAT

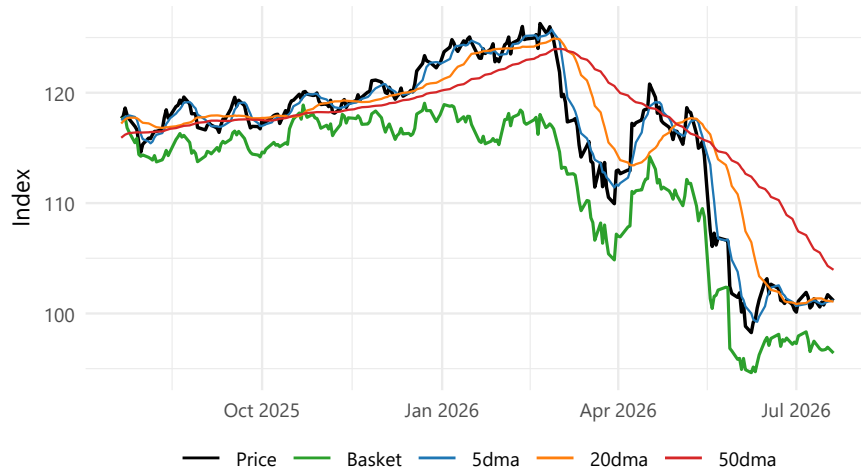
Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-07-13

- Current 252d return: -14.2% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4680
Malaysia Equity	1	0.1125
Indonesia Equity	1	0.1039
Philippines Equity	1	0.1011
Vietnam Equity	1	0.0748
Turkey Equity	1	0.0612
U.S. Dividend Equity	1	0.0549
Platinum	1	0.0236

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
NOBL	0.081	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1573	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1557	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1549	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1125	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
6	EIDO	0.1039	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1011	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0748	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0612	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0549	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0236	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.905 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: +0.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.269	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.760	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

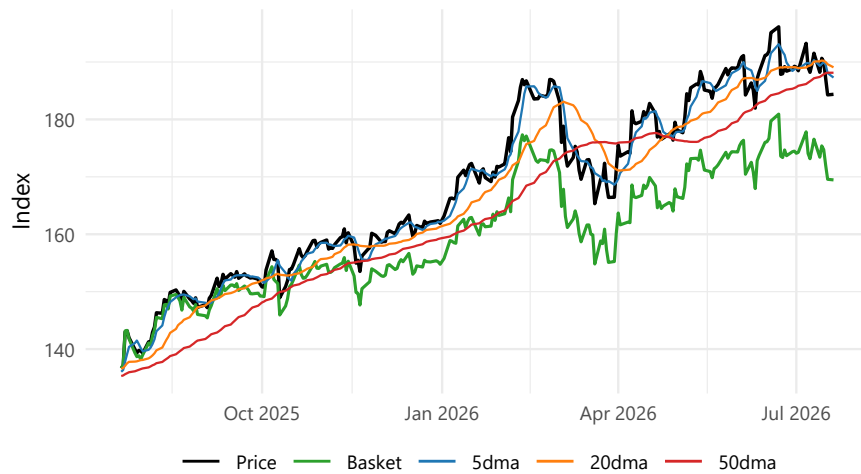
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0990	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0885	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +34.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-17 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7804

Asia-Pacific Equity	2	0.2196
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.043

ticker	rep_wt	name	category	etf_signal
EWJ	0.91	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

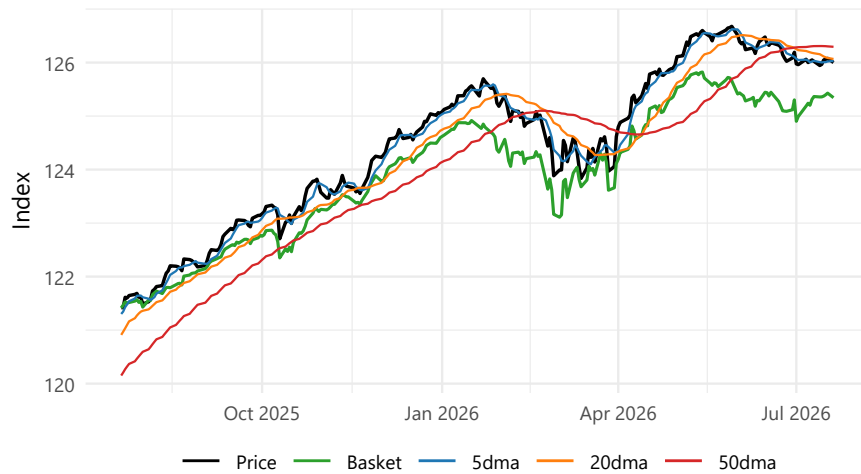
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	FLAT
6	DBJP	0.1117	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	FLAT
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1096	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1083	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1078	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.557 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +3.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3623
Interest Rate Hedged Fixed Income	3	0.3243
Floating Rate Fixed Income	3	0.3134

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	LONG
BKLN	0.214	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.302	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	FLAT
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	LONG
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	FLAT

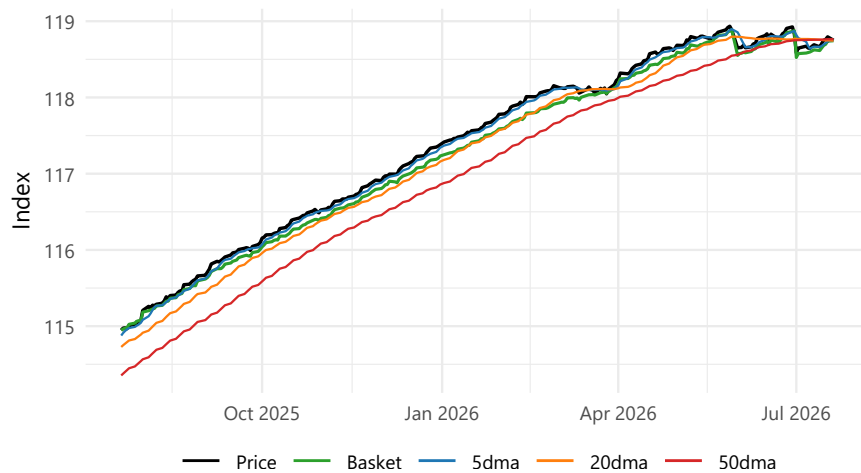
3	BKLN	0.1209	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1164	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1143	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1084	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	LONG
6	IGBH	0.1056	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1044	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	LONG
9	FLTR	0.1011	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.456 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +3.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.85) basket lead at lag 1: 0.065

ticker	rep_wt	name	category	etf_signal
SHV	0.275	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	LONG
MINT	0.172	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.122	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.403	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1310	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	LONG
6	ICSH	0.1297	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1162	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
2	MINT	0.1128	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1125	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1106	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.1019	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.0953	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0901	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (34 days, -2.4%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: -2.7% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.025

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1650	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1447	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.626 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-07-06
- Current 252d return: -31.3% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-06 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.024

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
INDS	cluster_16	Real Estate	2026-06-08	28	6.6%	0.00	1.00
RWO	cluster_16	Real Estate	2026-06-25	16	3.1%	0.00	1.00
IYR	cluster_16	Real Estate	2026-06-25	16	2.4%	0.01	0.99
FREL	cluster_16	Real Estate	2026-06-25	16	2.3%	0.01	0.99
XLRE	cluster_16	Real Estate	2026-06-25	16	1.4%	0.07	0.90
NURE	cluster_16	Real Estate	2026-04-13	67	13.6%	0.48	0.52
DIV	cluster_7	U.S. Low Volatility Equ...	2026-06-29	14	2.1%	0.00	1.00
SPHD	cluster_7	U.S. Low Volatility Equ...	2026-06-08	28	4.9%	0.00	1.00
FVD	cluster_7	U.S. Low Volatility Equ...	2026-06-10	26	3.3%	0.01	0.99
HDV	cluster_7	U.S. Low Volatility Equ...	2026-06-25	16	2.4%	0.01	0.99
FTCS	cluster_7	U.S. Low Volatility Equ...	2026-06-26	15	2.6%	0.01	0.99
FDL	cluster_7	U.S. Low Volatility Equ...	2026-07-16	2	-0.8%	0.00	0.98
PBJ	cluster_7	U.S. Low Volatility Equ...	2026-07-13	5	-0.1%	0.01	0.98
LGLV	cluster_7	U.S. Low Volatility Equ...	2026-06-26	15	1.4%	0.03	0.97
PSCC	cluster_7	U.S. Low Volatility Equ...	2026-06-10	26	6.4%	0.05	0.95
SPLV	cluster_7	U.S. Low Volatility Equ...	2026-06-11	25	3.0%	0.11	0.89
LVHD	cluster_7	U.S. Low Volatility Equ...	2026-06-10	26	2.8%	0.14	0.86
IYK	cluster_7	U.S. Low Volatility Equ...	2026-06-09	27	2.7%	0.19	0.81
SDY	cluster_7	U.S. Low Volatility Equ...	2026-05-20	40	4.0%	0.26	0.74
ACWV	cluster_7	U.S. Low Volatility Equ...	2026-07-06	10	-0.4%	0.09	0.70
KXI	cluster_7	U.S. Low Volatility Equ...	2026-07-16	2	-0.8%	0.16	0.67
IDU	cluster_7	U.S. Low Volatility Equ...	2026-06-25	16	-1.5%	0.54	0.46
USMV	cluster_7	U.S. Low Volatility Equ...	2026-06-29	14	0.4%	0.40	0.44
XLP	cluster_7	U.S. Low Volatility Equ...	2026-07-16	2	-1.1%	0.30	0.31
FUTY	cluster_7	U.S. Low Volatility Equ...	2026-06-25	16	-2.0%	0.73	0.25
XLU	cluster_7	U.S. Low Volatility Equ...	2026-06-25	16	-2.0%	0.78	0.21
FSTA	cluster_7	U.S. Low Volatility Equ...	2026-07-16	2	-1.2%	0.40	0.20
VPU	cluster_7	U.S. Low Volatility Equ...	2026-06-25	16	-2.0%	0.80	0.18
VDC	cluster_7	U.S. Low Volatility Equ...	2026-07-16	2	-1.3%	0.44	0.14
FXO	cluster_2	U.S. Large Cap Value	2026-06-05	29	8.6%	0.00	1.00
FNCL	cluster_2	U.S. Large Cap Value	2026-04-13	67	8.7%	0.00	1.00
XLF	cluster_2	U.S. Large Cap Value	2026-04-13	67	8.5%	0.00	1.00
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	73	12.2%	0.00	1.00
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	67	6.1%	0.00	1.00
FTA	cluster_2	U.S. Large Cap Value	2026-05-21	39	5.2%	0.00	1.00
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	67	6.0%	0.01	0.99
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	68	12.2%	0.01	0.99

IAT	cluster_2	U.S. Large Cap Value	2026-05-20	40	11.7%	0.01	0.99
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	68	11.3%	0.01	0.99
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	68	9.6%	0.01	0.99
PKW	cluster_2	U.S. Large Cap Value	2026-05-20	40	5.5%	0.01	0.99
KCE	cluster_2	U.S. Large Cap Value	2026-07-07	9	0.3%	0.01	0.99
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	69	8.8%	0.02	0.98
IAI	cluster_2	U.S. Large Cap Value	2026-07-06	10	-0.8%	0.02	0.98
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	68	11.2%	0.02	0.98
SCHD	cluster_2	U.S. Large Cap Value	2026-07-07	9	0.6%	0.00	0.97
KIE	cluster_2	U.S. Large Cap Value	2026-06-10	26	10.4%	0.05	0.95
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	69	9.6%	0.09	0.91
SIZE	cluster_2	U.S. Large Cap Value	2026-04-13	67	6.6%	0.27	0.73
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	66	6.4%	0.40	0.60
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	68	8.8%	0.41	0.59
IMCB	cluster_2	U.S. Large Cap Value	2026-04-10	68	10.8%	0.69	0.31
IWR	cluster_2	U.S. Large Cap Value	2026-04-10	68	8.1%	0.85	0.15
VO	cluster_2	U.S. Large Cap Value	2026-04-13	67	6.6%	0.90	0.10
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	67	7.1%	1.00	0.00
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	69	6.3%	1.00	-0.91
EWS	cluster_9	Asia-Pacific Equity	2026-06-17	21	6.0%	0.00	1.00
ENZL	cluster_9	Asia-Pacific Equity	2026-07-07	9	2.8%	0.00	1.00
EWZ	cluster_9	Asia-Pacific Equity	2026-07-16	2	0.4%	0.00	1.00
IDLV	cluster_9	Asia-Pacific Equity	2026-07-10	6	1.0%	0.00	1.00
EWX	cluster_9	Asia-Pacific Equity	2026-07-10	6	-0.1%	0.00	1.00
BBAX	cluster_9	Asia-Pacific Equity	2026-07-14	4	-0.2%	0.00	1.00
EPP	cluster_9	Asia-Pacific Equity	2026-07-15	3	-1.1%	0.00	1.00
EFAV	cluster_9	Asia-Pacific Equity	2026-07-16	2	0.4%	0.00	0.99
DWX	cluster_9	Asia-Pacific Equity	2026-07-09	7	1.4%	0.00	0.97
RWX	cluster_9	Asia-Pacific Equity	2026-07-16	2	0.0%	0.01	0.96
VNQI	cluster_9	Asia-Pacific Equity	2026-07-20	0	0.0%	0.02	0.86
PID	cluster_9	Asia-Pacific Equity	2026-07-16	2	-0.7%	0.00	0.78
EWA	cluster_9	Asia-Pacific Equity	2026-07-20	0	0.0%	0.00	0.73
EWL	cluster_9	Asia-Pacific Equity	2026-06-25	16	-0.7%	0.34	0.61
NFRA	cluster_9	Asia-Pacific Equity	2026-07-16	2	-0.4%	0.01	0.55
IGF	cluster_9	Asia-Pacific Equity	2026-07-09	7	-0.1%	0.37	0.50
EBND	cluster_9	Asia-Pacific Equity	2026-07-15	3	-0.5%	0.81	-0.57
EMLC	cluster_9	Asia-Pacific Equity	2026-07-15	3	-0.7%	0.88	-0.64
DBA	cluster_19	Broad Commodities	2026-07-09	7	1.1%	0.00	1.00
FTGC	cluster_19	Broad Commodities	2026-07-17	1	0.2%	0.00	0.97
DBE	cluster_19	Broad Commodities	2026-07-20	0	0.0%	0.00	0.90
DBC	cluster_19	Broad Commodities	2026-06-03	31	-3.8%	0.00	0.73
DJP	cluster_19	Broad Commodities	2026-06-03	31	-3.9%	0.00	0.72
PDBC	cluster_19	Broad Commodities	2026-06-03	31	-3.7%	0.00	0.69
USO	cluster_19	Broad Commodities	2026-06-03	31	-10.9%	0.00	0.44
XHE	cluster_13	Healthcare	2026-06-09	27	4.8%	0.09	0.91

RSPH	cluster_13	Healthcare	2026-05-21	39	9.3%	0.24	0.76
IHF	cluster_13	Healthcare	2026-05-08	48	13.0%	0.27	0.73
PSCH	cluster_13	Healthcare	2026-04-13	67	23.7%	0.34	0.66
XLV	cluster_13	Healthcare	2026-05-21	39	7.5%	0.44	0.56
FHLC	cluster_13	Healthcare	2026-05-19	41	9.3%	0.45	0.55
VHT	cluster_13	Healthcare	2026-05-19	41	9.3%	0.48	0.52
IYH	cluster_13	Healthcare	2026-05-20	40	8.2%	0.49	0.51
FXH	cluster_13	Healthcare	2026-04-10	68	9.2%	0.50	0.50
PTH	cluster_13	Healthcare	2026-06-17	21	10.3%	0.56	0.44
PPH	cluster_13	Healthcare	2026-05-21	39	4.0%	0.57	0.42
IBB	cluster_13	Healthcare	2026-06-17	21	7.5%	0.63	0.37
OUSA	cluster_5	U.S. Large Cap Equity	2026-07-01	12	1.4%	0.02	0.97
DGRO	cluster_5	U.S. Large Cap Equity	2026-04-13	67	6.1%	0.09	0.91
GSLC	cluster_5	U.S. Large Cap Equity	2026-06-30	13	-0.3%	0.09	0.70
VIG	cluster_5	U.S. Large Cap Equity	2026-04-14	66	5.4%	0.40	0.60
SPYX	cluster_5	U.S. Large Cap Equity	2026-06-30	13	-0.3%	0.13	0.57
OEF	cluster_5	U.S. Large Cap Equity	2026-07-06	10	-0.7%	0.03	0.53
IVV	cluster_5	U.S. Large Cap Equity	2026-06-30	13	-0.5%	0.14	0.46
URTH	cluster_5	U.S. Large Cap Equity	2026-07-01	12	-0.5%	0.16	0.42
CATH	cluster_5	U.S. Large Cap Equity	2026-07-06	10	-1.0%	0.10	0.40
SPY	cluster_5	U.S. Large Cap Equity	2026-06-30	13	-0.6%	0.16	0.39
DGRW	cluster_5	U.S. Large Cap Equity	2026-07-06	10	-0.7%	0.11	0.37
VOO	cluster_5	U.S. Large Cap Equity	2026-06-30	13	-0.7%	0.18	0.35
IWL	cluster_5	U.S. Large Cap Equity	2026-07-01	12	-0.6%	0.14	0.34
IOO	cluster_5	U.S. Large Cap Equity	2026-07-15	3	-1.9%	0.01	0.32
MGC	cluster_5	U.S. Large Cap Equity	2026-07-01	12	-0.6%	0.16	0.27
DIA	cluster_5	U.S. Large Cap Equity	2026-04-14	66	6.7%	0.74	0.26
CRBN	cluster_5	U.S. Large Cap Equity	2026-06-30	13	-1.5%	0.54	-0.40
XMLV	cluster_1	U.S. Mid Cap Equity	2026-06-25	16	3.2%	0.00	1.00
REM	cluster_1	U.S. Mid Cap Equity	2026-07-09	7	-0.6%	0.00	1.00
SMMV	cluster_1	U.S. Mid Cap Equity	2026-06-25	16	3.6%	0.01	0.99
REGL	cluster_1	U.S. Mid Cap Equity	2026-06-12	24	2.7%	0.03	0.97
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	35	3.9%	0.07	0.93
RTH	cluster_1	U.S. Mid Cap Equity	2026-07-17	1	-0.5%	0.00	0.88
IJS	cluster_1	U.S. Mid Cap Equity	2026-04-10	68	10.7%	0.16	0.84
SDVY	cluster_1	U.S. Mid Cap Equity	2026-06-08	28	4.1%	0.21	0.79
CGW	cluster_1	U.S. Mid Cap Equity	2026-06-25	16	-0.2%	0.21	0.79
ISMD	cluster_1	U.S. Mid Cap Equity	2026-04-09	69	16.3%	0.41	0.59
PSCD	cluster_1	U.S. Mid Cap Equity	2026-05-27	36	7.6%	0.49	0.51
PHO	cluster_1	U.S. Mid Cap Equity	2026-06-26	15	-1.6%	0.52	0.48
FIW	cluster_1	U.S. Mid Cap Equity	2026-06-16	22	0.2%	0.61	0.39
FTXR	cluster_1	U.S. Mid Cap Equity	2026-04-17	63	7.3%	0.71	0.25
VCR	cluster_1	U.S. Mid Cap Equity	2026-07-01	12	-2.5%	0.40	0.15
FDIS	cluster_1	U.S. Mid Cap Equity	2026-07-01	12	-2.5%	0.40	0.14
IYC	cluster_1	U.S. Mid Cap Equity	2026-07-06	10	-2.2%	0.57	-0.31

XLY	cluster_1	U.S. Mid Cap Equity	2026-07-06	10	-2.9%	0.48	-0.34
KURE	cluster_14	China Equity	2026-07-06	10	2.0%	0.00	1.00
FXB	cluster_25	International Fixed Inc...	2026-07-15	3	-0.7%	0.00	0.99
GSY	cluster_24	Cash / Ultra-Short Fixe...	2026-07-07	9	-0.2%	0.19	0.48
SHV	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	0	0.0%	0.40	0.07
BIL	cluster_24	Cash / Ultra-Short Fixe...	2026-07-16	2	0.0%	0.47	0.01
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	0	0.0%	0.45	-0.10
MINT	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	0	0.0%	0.70	-0.39
AML	cluster_10	Materials	2026-07-10	6	3.4%	0.00	1.00
MLPA	cluster_10	Materials	2026-07-10	6	3.1%	0.00	1.00
MOO	cluster_10	Materials	2026-07-07	9	1.4%	0.00	1.00
MLPX	cluster_10	Materials	2026-07-09	7	1.7%	0.00	1.00
EMLP	cluster_10	Materials	2026-07-08	8	0.2%	0.02	0.98
RSPM	cluster_10	Materials	2026-06-16	22	-4.7%	0.94	-0.94
EUFN	cluster_4	Developed ex-US Equity	2026-06-12	24	2.2%	0.00	1.00
HDEF	cluster_4	Developed ex-US Equity	2026-07-09	7	1.2%	0.00	0.99
VIGI	cluster_4	Developed ex-US Equity	2026-06-30	13	1.0%	0.01	0.99
EFV	cluster_4	Developed ex-US Equity	2026-07-08	8	0.4%	0.00	0.98
EPOL	cluster_4	Developed ex-US Equity	2026-07-08	8	1.7%	0.00	0.97
VYMI	cluster_4	Developed ex-US Equity	2026-07-08	8	0.7%	0.01	0.95
IPKW	cluster_4	Developed ex-US Equity	2026-07-10	6	-0.1%	0.00	0.93
QEFA	cluster_4	Developed ex-US Equity	2026-07-06	10	-1.3%	0.02	0.91
EWU	cluster_4	Developed ex-US Equity	2026-07-16	2	-1.2%	0.05	0.80
GSIE	cluster_4	Developed ex-US Equity	2026-07-02	11	-0.7%	0.24	0.55
IQLT	cluster_4	Developed ex-US Equity	2026-07-01	12	0.1%	0.25	0.52
RODM	cluster_4	Developed ex-US Equity	2026-07-20	0	0.0%	0.02	0.41
INTF	cluster_4	Developed ex-US Equity	2026-07-15	3	-1.9%	0.16	0.27
IEV	cluster_4	Developed ex-US Equity	2026-07-01	12	-0.3%	0.39	0.16
IEUR	cluster_4	Developed ex-US Equity	2026-07-15	3	-1.7%	0.29	0.15
IVLU	cluster_4	Developed ex-US Equity	2026-07-15	3	-2.0%	0.15	0.06
ESGD	cluster_4	Developed ex-US Equity	2026-06-30	13	-1.1%	0.43	0.04
VGK	cluster_4	Developed ex-US Equity	2026-07-01	12	-0.1%	0.43	0.02
EWJ	cluster_4	Developed ex-US Equity	2026-04-10	68	8.0%	0.98	-0.14
EFA	cluster_4	Developed ex-US Equity	2026-07-01	12	-0.4%	0.50	-0.23
IEFA	cluster_4	Developed ex-US Equity	2026-07-15	3	-2.0%	0.44	-0.25
IQDF	cluster_4	Developed ex-US Equity	2026-07-15	3	-2.1%	0.60	-0.54
FLOT	cluster_23	Bank Loan / Floating Ra...	2026-07-10	6	0.0%	0.65	-0.40
FLRN	cluster_23	Bank Loan / Floating Ra...	2026-07-13	5	0.0%	0.72	-0.48
EWJV	cluster_22	Japan Equity	2026-07-06	10	-2.6%	0.09	0.68
IPAC	cluster_22	Japan Equity	2026-06-30	13	-1.2%	0.75	-0.63
SLV	cluster_18	Gold	2026-05-27	36	-24.5%	0.86	-0.86
SIVR	cluster_18	Gold	2026-05-27	36	-24.4%	0.86	-0.86
BSCQ	cluster_6	Aggregate Fixed Income	2026-07-09	7	-0.2%	0.19	0.36
IBUY	cluster_3	U.S. Large Cap Growth	2026-06-15	23	4.5%	0.00	1.00
FDN	cluster_3	U.S. Large Cap Growth	2026-07-07	9	-0.4%	0.01	0.86

SCHG	cluster_3	U.S. Large Cap Growth	2026-07-06	10	-1.1%	0.02	0.84
VOX	cluster_3	U.S. Large Cap Growth	2026-07-15	3	-3.2%	0.00	0.73
FCOM	cluster_3	U.S. Large Cap Growth	2026-07-15	3	-3.2%	0.00	0.72
SKYY	cluster_3	U.S. Large Cap Growth	2026-07-01	12	0.3%	0.11	0.56
MGK	cluster_3	U.S. Large Cap Growth	2026-07-06	10	-1.7%	0.12	0.17
VUG	cluster_3	U.S. Large Cap Growth	2026-07-10	6	-2.4%	0.11	0.15
ARKW	cluster_3	U.S. Large Cap Growth	2026-07-02	11	-1.6%	0.38	0.02
VOOG	cluster_3	U.S. Large Cap Growth	2026-07-01	12	-1.5%	0.32	-0.13
IVW	cluster_3	U.S. Large Cap Growth	2026-07-01	12	-1.6%	0.32	-0.14
SPYG	cluster_3	U.S. Large Cap Growth	2026-07-01	12	-1.6%	0.33	-0.15
IUSG	cluster_3	U.S. Large Cap Growth	2026-07-01	12	-1.7%	0.36	-0.19
ONEQ	cluster_3	U.S. Large Cap Growth	2026-07-15	3	-2.8%	0.49	-0.46
ILCG	cluster_3	U.S. Large Cap Growth	2026-06-30	13	-3.9%	0.67	-0.65
PTF	cluster_3	U.S. Large Cap Growth	2026-04-07	71	8.0%	1.00	-1.00
QTUM	cluster_3	U.S. Large Cap Growth	2026-04-10	68	21.0%	1.00	-1.00
HYLS	cluster_11	High Yield Bond	2026-07-20	0	0.0%	0.19	0.33
VRP	cluster_11	High Yield Bond	2026-07-02	11	-0.5%	0.55	-0.12
SCHE	cluster_8	Emerging Markets Equity	2026-07-14	4	-1.6%	0.88	-0.84
AIA	cluster_8	Emerging Markets Equity	2026-07-10	6	-5.0%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
EWS	cluster_9	Asia-Pacific Equity	1.00	1.00
AML	cluster_10	Materials	1.00	1.00
KURE	cluster_14	China Equity	1.00	1.00
DBA	cluster_19	Broad Commodities	1.00	1.00
IPAY	cluster_3	U.S. Large Cap Growth	1.00	1.00
FXO	cluster_2	U.S. Large Cap Value	1.00	1.00
DIV	cluster_7	U.S. Low Volatility Equity	1.00	1.00
XMLV	cluster_1	U.S. Mid Cap Equity	1.00	1.00
INDS	cluster_16	Real Estate	1.00	1.00
EUFN	cluster_4	Developed ex-US Equity	1.00	1.00
QDF	cluster_5	U.S. Large Cap Equity	1.00	0.98
SDIV	cluster_8	Emerging Markets Equity	0.92	0.92
XHE	cluster_13	Healthcare	1.00	0.91
EWJV	cluster_22	Japan Equity	0.77	0.68
FTSL	cluster_23	Bank Loan / Floating Rate	0.52	0.49
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.67	0.48
BSCQ	cluster_6	Aggregate Fixed Income	0.55	0.36
HYLS	cluster_11	High Yield Bond	0.52	0.33
IMCG	cluster_12	U.S. Momentum Equity	0.84	-0.16
SGOL	cluster_18	Gold	0.00	-0.69

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
QTUM	cluster_3	U.S. Large Cap Growth	1.00	-1.00
PCY	cluster_11	High Yield Bond	1.00	-1.00
IGBH	cluster_23	Bank Loan / Floating Rate	1.00	-1.00
URA	cluster_10	Materials	1.00	-1.00
XMMO	cluster_12	U.S. Momentum Equity	1.00	-1.00
EWY	cluster_8	Emerging Markets Equity	1.00	-1.00

UNG	cluster_19	Broad Commodities	1.00	-1.00
CNXT	cluster_14	China Equity	1.00	-1.00
VNM	cluster_20	India Equity	1.00	-1.00
ITM	cluster_17	Municipal Bond	1.00	-1.00
SPLB	cluster_15	Fixed Income	1.00	-1.00
VPL	cluster_22	Japan Equity	1.00	-1.00
LTPZ	cluster_21	Inflation-Protected Fixed...	1.00	-1.00
AGGY	cluster_6	Aggregate Fixed Income	1.00	-1.00
VLUE	cluster_2	U.S. Large Cap Value	1.00	-1.00
HFXI	cluster_4	Developed ex-US Equity	1.00	-1.00
QAI	cluster_5	U.S. Large Cap Equity	1.00	-1.00
IJK	cluster_1	U.S. Mid Cap Equity	1.00	-1.00
GOAU	cluster_18	Gold	0.99	-0.99
ECH	cluster_9	Asia-Pacific Equity	0.93	-0.93

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.303	0.094	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.484	0.075	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.737	0.036	0.00008	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.632	0.214	0.00036	
FXF	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.731	0.097	0.00023	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	LONG	0.592	0.170	0.00040	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.740	0.126	0.00031	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.740	0.153	0.00040	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.698	0.191	0.00046	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.678	0.180	0.00047	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.731	0.181	0.00045	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	FLAT	0.507	0.219	0.00055	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.867	0.020	0.00001	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_6	43	FLAT	0.847	0.127	0.00037	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.867	0.073	0.00021	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.818	0.203	0.00049	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_6	43	FLAT	0.877	0.164	0.00032	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_6	43	FLAT	0.836	0.239	0.00048	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.446	0.274	0.00074	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_17	13	FLAT	0.878	0.249	0.00046	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.446	0.330	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_14	15	LONG	0.789	0.270	0.00071	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_17	13	FLAT	0.878	0.264	0.00054	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_10	21	FLAT	0.746	0.357	0.00084	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.503	0.353	0.00101	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.866	0.327	0.00071	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_14	15	FLAT	0.855	0.255	0.00072	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.558	0.406	0.00106	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_14	15	FLAT	0.903	0.238	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.919	0.232	0.00058	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20